Summary in Graph

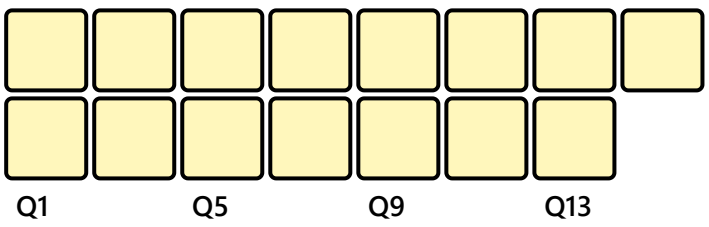
# Exam Summary (GO Classes CS | CO and Architecture | Cache Memory | Weekly Quiz 1)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 15 (5 + 10) |
| Total Marks: | 25 (5 + 20) |
| Exam Duration: | 45 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (36) out of 355 users

EXAM RESPONSE EXAM STATS FEEDBACK

## Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 CO & Architecture

Which cache mapping function does not require a replacement algorithm?

A. Direct mapping

B. Set associative mapping

C. Fully associative mapping

D. None

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which cache mapping function is least likely to thrash, i.e., it has the lowest chance of two blocks contending with each other to be stored in the same line?

A. Direct mapping

B. Set associative mapping

C. Fully associative mapping

D. All are equally likely

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which is the fastest cache mapping function? ("Fastest" in terms of access/lookup speed)

A. Direct mapping

B. Set associative mapping

C. Fully associative mapping

D. All are equally fast

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a $4$-way set associative cache ( $\mathrm{E} = 4$ ). Which one of the following statements is true?

A. The cache has $4$ blocks per line.

B. The cache has $4$ sets per line.

C. The cache has $4$ lines per set.

D. The cache has $4$ sets per block.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Which one of the following statements about cache memories is true?

A. Fully associative caches offer better latency, while direct-mapped caches have lower miss rates.

B. Fully associative caches offer lower miss rates, while direct-mapped caches have better latency.

C. Direct-mapped caches have both better miss rates and better latency.

D. Both generally have similar latency and miss rates.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following cache represents a 2-way set associative cache, i.e., there are two lines per set. Notice that the set ID values start at 011011012 and increment every other row. This is meant to imply that you are looking at a group of lines/sets toward the middle of the cache and not the entire cache. There are 14 bits for the tag, 8 bits for the set ID, and 2 bits for the word ID.

| Tag (binary values) | Set ID (binary values) | Word within block | | | |
|---|---|---|---|---|---|
| | | 00 | 01 | 10 | 11 |
| 10100001001001 | 01101101 | $00_{16}$ | $61_{16}$ | $C2_{16}$ | $23_{16}$ |
| 11100001100100 | 01101101 | $10_{16}$ | $71_{16}$ | $D2_{16}$ | $33_{16}$ |
| 11001011010110 | 01101110 | $20_{16}$ | $81_{16}$ | $E2_{16}$ | $43_{16}$ |
| 11100101101011 | 01101110 | $30_{16}$ | $91_{16}$ | $F2_{16}$ | $53_{16}$ |
| 11110110110100 | 01101111 | $40_{16}$ | $A1_{16}$ | $02_{16}$ | $63_{16}$ |
| 10100111010101 | 01101111 | $50_{16}$ | $B1_{16}$ | $12_{16}$ | $73_{16}$ |
| 10101010111110 | 01110000 | $84_{16}$ | $E5_{16}$ | $46_{16}$ | $A7_{16}$ |
| 10101010010011 | 01110000 | $94_{16}$ | $F5_{16}$ | $56_{16}$ | $B7_{16}$ |
| 01110001001000 | 01110001 | $A4_{16}$ | $A5_{16}$ | $66_{16}$ | $C7_{16}$ |
| 00001101101101 | 01110001 | $B4_{16}$ | $15_{16}$ | $76_{16}$ | $D7_{16}$ |
| 01011010010010 | 01110010 | $C4_{16}$ | $25_{16}$ | $86_{16}$ | $E7_{16}$ |
| 10101111001011 | 01110010 | $D4_{16}$ | $35_{16}$ | $96_{16}$ | $F7_{16}$ |

How many lines are contained in this cache?

Your Answer: | Correct Answer: 512 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following cache represents a $2$-way set associative cache, i.e., there are two lines per set. Notice that the set ID values start at $011011012$ and increment every other row. This is meant to imply that you are looking at a group of lines/sets toward the middle of the cache and not the entire cache. There are $14$ bits for the tag, $8$ bits for the set id, and $2$ bits for the word id.

| Tag (binary values) | Set ID (binary values) | Word within block | | | |
|---|---|---|---|---|---|
| | | 00 | 01 | 10 | 11 |
| 10100001001001 | 01101101 | $00_{16}$ | $61_{16}$ | $C2_{16}$ | $23_{16}$ |
| 11100001100100 | 01101101 | $10_{16}$ | $71_{16}$ | $D2_{16}$ | $33_{16}$ |
| 11001011010110 | 01101110 | $20_{16}$ | $81_{16}$ | $E2_{16}$ | $43_{16}$ |
| 11100101101011 | 01101110 | $30_{16}$ | $91_{16}$ | $F2_{16}$ | $53_{16}$ |
| 11110110110100 | 01101111 | $40_{16}$ | $A1_{16}$ | $02_{16}$ | $63_{16}$ |
| 10100111010101 | 01101111 | $50_{16}$ | $B1_{16}$ | $12_{16}$ | $73_{16}$ |
| 10101010111110 | 01110000 | $84_{16}$ | $E5_{16}$ | $46_{16}$ | $A7_{16}$ |
| 10101010010011 | 01110000 | $94_{16}$ | $F5_{16}$ | $56_{16}$ | $B7_{16}$ |
| 01110001001000 | 01110001 | $A4_{16}$ | $A5_{16}$ | $66_{16}$ | $C7_{16}$ |
| 00001101101101 | 01110001 | $B4_{16}$ | $15_{16}$ | $76_{16}$ | $D7_{16}$ |
| 01011010010010 | 01110010 | $C4_{16}$ | $25_{16}$ | $86_{16}$ | $E7_{16}$ |
| 10101111001011 | 01110010 | $D4_{16}$ | $35_{16}$ | $96_{16}$ | $F7_{16}$ |

If the number of blocks contained in the memory space (not the cache, but the memory) of the cache system defined above is $2^n$, then what is $n$ ?

Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #8 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The following cache represents a $2$-way set associative cache, i.e., there are two lines per set. Notice that the set ID values start at $011011012$ and increment every other row. This is meant to imply that you are looking at a group of lines/sets toward the middle of the cache and not the entire cache. There are $14$ bits for the tag, $8$ bits for the set id, and $2$ bits for the word id.

| Tag (binary values) | Set ID (binary values) | Word within block | | | |
|---|---|---|---|---|---|
| | | 00 | 01 | 10 | 11 |
| 10100001001001 | 01101101 | $00_{16}$ | $61_{16}$ | $C2_{16}$ | $23_{16}$ |
| 11100001100100 | 01101101 | $10_{16}$ | $71_{16}$ | $D2_{16}$ | $33_{16}$ |
| 11001011010110 | 01101110 | $20_{16}$ | $81_{16}$ | $E2_{16}$ | $43_{16}$ |
| 11100101101011 | 01101110 | $30_{16}$ | $91_{16}$ | $F2_{16}$ | $53_{16}$ |
| 11110110110100 | 01101111 | $40_{16}$ | $A1_{16}$ | $02_{16}$ | $63_{16}$ |
| 10100111010101 | 01101111 | $50_{16}$ | $B1_{16}$ | $12_{16}$ | $73_{16}$ |
| 10101010111110 | 01110000 | $84_{16}$ | $E5_{16}$ | $46_{16}$ | $A7_{16}$ |
| 10101010010011 | 01110000 | $94_{16}$ | $F5_{16}$ | $56_{16}$ | $B7_{16}$ |
| 01110001001000 | 01110001 | $A4_{16}$ | $A5_{16}$ | $66_{16}$ | $C7_{16}$ |
| 00001101101101 | 01110001 | $B4_{16}$ | $15_{16}$ | $76_{16}$ | $D7_{16}$ |
| 01011010010010 | 01110010 | $C4_{16}$ | $25_{16}$ | $86_{16}$ | $E7_{16}$ |
| 10101111001011 | 01110010 | $D4_{16}$ | $35_{16}$ | $96_{16}$ | $F7_{16}$ |

A copy of the data from memory address $7121C5$ (hex) is contained in the portion of the cache shown above. Enter the value that was retrieved from that address as an eight-bit binary number with no base identification, e.g., $0x4F$ (hexadecimal 4F) should be entered as $01001111$.

Your Answer: | Correct Answer: 10100101 | Not Attempted | Time taken: 00min 00sec | Discuss

## Q #9 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a $256$ KB $4$-way set associative cache with $256$ byte cache lines for a processor that uses $64$-bit data words and $48$-bit byte addresses. What is the size (in bits) of the tag field per cache block?

Your Answer: | Correct Answer: 32 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a $256$ KB $4$-way set associative cache with $256$ byte cache lines for a processor that uses $64$-bit data words and $48$-bit byte addresses. Assume the variable x, of type uint64_t, is stored in memory at location 0x4A85 B413 A518. (uint64_t means an unsigned $64$-bit integer)
Assume that x is present in the cache, and char* ptr $= 0 \times 4$ $\mathrm{A}85$ B400 0000. (Note: sizeof(char) = 1)

Consider the accesses in the following options. For each access, it can be a MISS or HIT or NOT ENOUGH INFO. For each access, it is a MISS if it "must" cause a miss, HIT if it will "never" cause a miss, or NOT ENOUGH INFO.
Determine which of the following accesses will cause a cache HIT.

A. *(ptr $+0 \times 11$ A538)

B. *(ptr $+0 \times 13$ A588)

C. *(ptr $+0 \times 13$ 0218)

D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Suppose a computer has an $8$-bit address space. It has a $2$-way set-associative cache with LRU replacement policy. Cache size is $64$ Bytes and block size is $8$ Bytes.
What is the length of the tag field (in bits) ?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Suppose a computer has an 8-bit address space. It has a $2$-way set-associative cache with LRU replacement policy. Cache size is $64$ Bytes and block size is $8$ Bytes. The addresses of memory access are as follows; the sequence is from the top to bottom in the table.

| Address | Hit or Miss |
|---|---|
| $0b10001000$ | Miss |
| $0b10011000$ | Miss |
| $0b10001001$ | Hit |
| $0b00001000$ | Miss |
| $0b00001001$ | |
| $0b10010001$ | |
| $0b11001001$ | |
| $0b00001010$ | |
| $0b10001001$ | |

What is the hit rate (in $\%$) of the memory accesses in the above table? (up to two decimal places)

Your Answer: | Correct Answer: 33.33 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Suppose a computer has an $8$-bit address space. It has a direct mapping cache. Cache size is $64$ Bytes and block size is $8$ Bytes. The addresses of memory access are as follows; the sequence is from the top to bottom in the table.

| Address | Hit or Miss |
|---|---|
| $0b10001000$ | Miss |
| $0b10011000$ | Miss |
| $0b10001001$ | Hit |
| $0b00001000$ | Miss |
| $0b00001001$ | |
| $0b10010001$ | |
| $0b11001001$ | |
| $0b00001010$ | |
| $0b10001001$ | |

What is the hit rate (in $\%$) of the memory accesses in the above table? (up to two decimal places)

Your Answer: | Correct Answer: 22.22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A direct-mapped cache has $64$ blocks and a block size of $16$ bytes. To what block number in the cache does byte address $1220_{10}$ map?

A. $11$

B. $13$

C. $12$

D. None of the other choices

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

The size of the physical address space of a processor is $2^P$ bytes. The word length is $2^W$ bytes. The capacity of cache memory is $2^N$ bytes. The size of each cache block is $2^M$ words. For a $K$-way set-associative cache memory, the length (in number of bits) of the tag field is

A. $P - N - \log_2 K$

B. $P - N + \log_2 K$

C. $P - N - M - W - \log_2 K$

D. $P - N - M - W + \log_2 K$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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